

# Lecture 18: Social media and society

Matthew J. Salganik

Sociology 204: Social Networks  
Princeton University



Social media:

- ▶ [Lecture 17: Social media and individuals](#)
- ▶ Lecture 18: Social media and society
- ▶ Lecture 19: Social media and democracy
- ▶ Lecture 20: Fixing social media

- ▶ Measuring and understanding the effect of social media on individuals is hard, in part because jingle-jangle problem and heterogeneity of social media and people

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- ▶ Being off FB for one month also leads to new awareness about the good and bad aspects of FB

Now it is your turn: self-experimentation and social media

## Self-experimentation and social media

- ▶ Assignment 5: Self-experimentation and social media, design

## Self-experimentation and social media

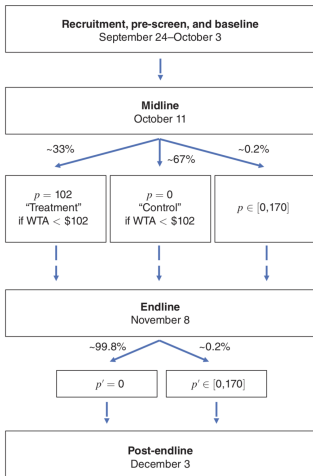
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- ▶ Assignment 6: Self-experimentation and social media, implement
- ▶ Assignment 7: Self-experimentation and social media, results
- ▶ Assignment 8: Self-experimentation and social media, reflect



Daily text messages  
September 27–November 8

# Surprises From Self-Experimentation: Sleep, Mood, and Weight

Seth Roberts

## A few notes on self-experimentation

- ▶ Long history in medicine: [https://en.wikipedia.org/wiki/Self-experimentation\\_in\\_medicine](https://en.wikipedia.org/wiki/Self-experimentation_in_medicine)

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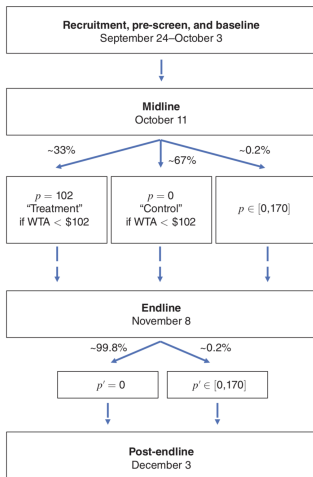
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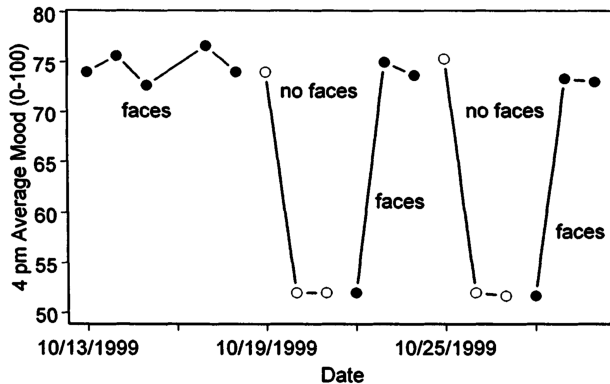
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- ▶ Can't learn about heterogeneity across people
- ▶ Can learn about unexpected outcomes
- ▶ Can iterate quickly



Daily text messages  
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You will be your own control group ("within-subjects design")

A few notes about this assignment:

- ▶ Choose an interesting and important intervention. Best ones will likely be either motivated by science or wellbeing.

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- ▶ If this assignment causes problems, come talk to us or University Health Services.

## Social media:

- ▶ Lecture 17: Social media and individuals
- ▶ [Lecture 18: Social media and society](#)
- ▶ Lecture 19: Social media and democracy
- ▶ Lecture 20: Fixing social media

1. Crockett (2017). Moral outrage in the digital age. *Nature Human Behavior*.
2. Brady et al. (2021). How social learning amplifies moral outrage expression in online social networks. *Science Advances*.
3. Aral (2018). How lies spread online. *New York Times*.
4. Vosoughi et al. (2018). The spread of true and false news online. *Science*.
5. William Brady (2024) Bluesky thread about paper: "Misinformation exploits outrage to spread online".
6. Molly Crockett (2024) Bluesky thread about creating the paper: "Misinformation exploits outrage to spread online".
7. Brady et al. (2024) Misinformation exploits outrage to spread online. *Science* 386(6725):991-996.

Background

**SOCIAL SCIENCE**

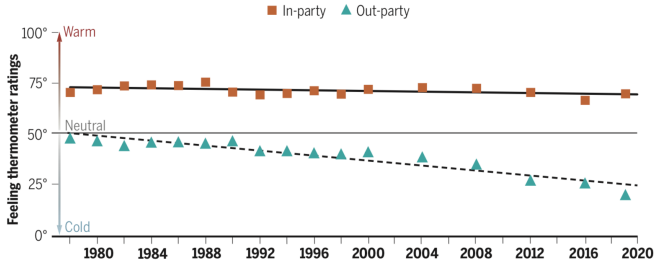
# Political sectarianism in America

A poisonous cocktail of othering, aversion, and moralization poses a threat to democracy

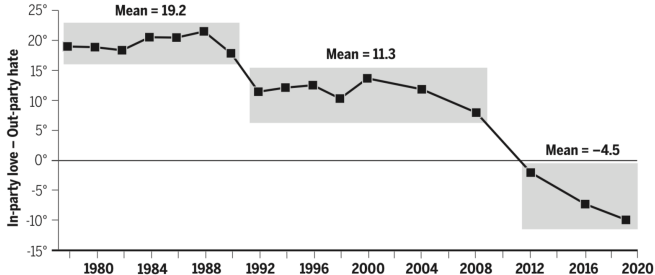
*By* **Eli J. Finkel<sup>1</sup>, Christopher A. Bail<sup>2</sup>, Mina Cikara<sup>3</sup>, Peter H. Ditto<sup>4</sup>, Shanto Iyengar<sup>5</sup>, Samara Klar<sup>6</sup>, Lilliana Mason<sup>7</sup>, Mary C. McGrath<sup>1</sup>, Brendan Nyhan<sup>8</sup>, David G. Rand<sup>9</sup>, Linda J. Skitka<sup>10</sup>, Joshua A. Tucker<sup>11</sup>, Jay J. Van Bavel<sup>11</sup>, Cynthia S. Wang<sup>1</sup>, James N. Druckman<sup>1</sup>**

Published in 2020

## Warmth toward the opposing party (out-party) has diminished for decades



## Out-party hate has emerged as a stronger force than in-party love

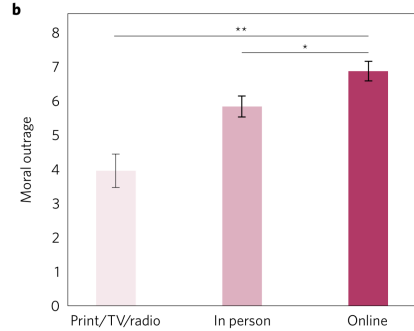
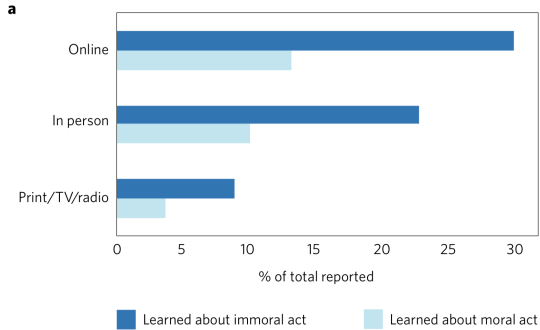


## The Filter Bubble

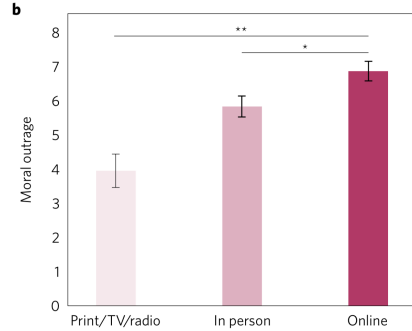
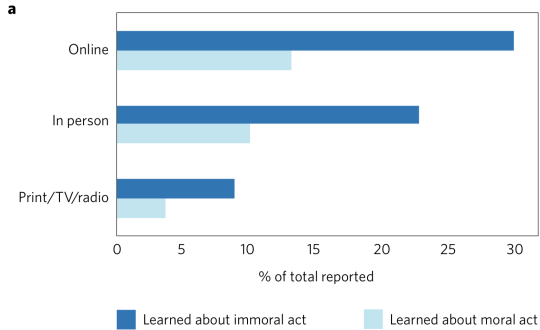
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Eli Pariser

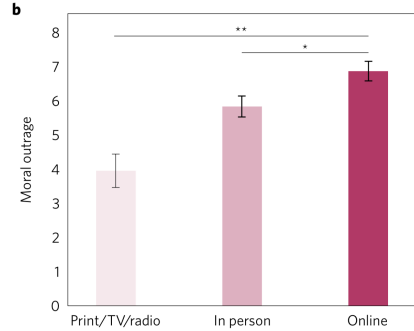
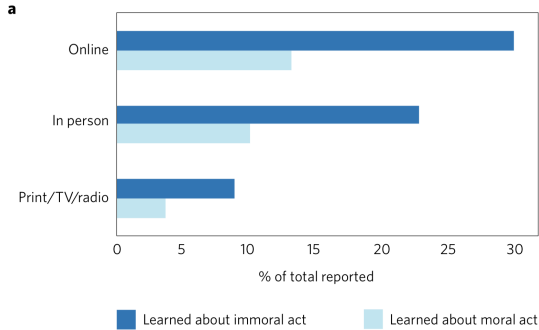
What goes viral? Moral outrage and lies



► People more likely to learn about immoral acts online than through other media



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- ▶ Imoral acts that people see online lead to more outrage



- ▶ People more likely to learn about immoral acts online than through other media
- ▶ Imoral acts that people see online lead to more outrage
- ▶ “Digital media transform moral outrage by changing both the nature and prevalence of the stimuli that trigger it.”

**SOCIAL SCIENCES**

# **How social learning amplifies moral outrage expression in online social networks**

**William J. Brady<sup>1\*</sup>, Killian McLoughlin<sup>1</sup>, Tuan N. Doan<sup>2</sup>, Molly J. Crockett<sup>1\*</sup>**

- ▶ two observational studies and two experiments (all preregistered)

## **SOCIAL SCIENCES**

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- ▶ two observational studies and two experiments (all preregistered)
- ▶ reinforcement learning and norm learning, both of which are controlled in part by design of social media platforms

“A person can be viewed as expressing moral outrage if:

- ▶ they have feelings in response to a perceived violation of their personal morals
- ▶ their feelings are comprised of emotions such as anger, disgust, and contempt
- ▶ the feelings are associated with specific reactions including blaming people/events/things, holding them responsible or wanting to punish them.”

They labeled a many example tweets and then built a Digital Outrage Classifier



<https://www.youtube.com/watch?v=b2AYlD8ReeA&t=11m26s>

Supporting our hypotheses, we found that daily outrage expression was significantly and positively associated with the amount of social feedback received for the previous day's outrage expression (Study 1:  $b = 0.03$ ,  $p < .001$ , 95% CI = [0.03, 0.03]; Study 2:  $b = 0.02$ ,  $p < .001$ , 95% CI = [0.02, 0.03]). For our model, this effect size translates to an expected 2-3% increase in outrage expression on the following day of tweeting if a user received a 100% increase in feedback for expressing outrage on a given day. For instance, a user who averaged 5 likes/shares per tweet, and then received 10 likes/shares when they expressed outrage, would be expected to increase their outrage expression on the next day by 2-3%. While this effect size is small, it can easily scale on social media over time, become notable at scale at the network level, or for users who maintain a larger followership and could experience much higher than 100% increases in social feedback for tweeting outrage content (e.g., political leaders). For other model specifications to test the robustness of the effect, see SOM, Section 2.0.

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- ▶ “outrage expression on Twitter can be explained in part by variation in social feedback that people receive via the platform”

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- ▶ “outrage expression on Twitter can be explained in part by variation in social feedback that people receive via the platform”
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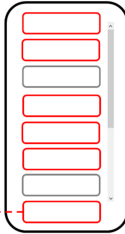
### Three main findings from observational studies

- ▶ “outrage expression on Twitter can be explained in part by variation in social feedback that people receive via the platform”
- ▶ “users are more likely to express outrage in more ideologically extreme social networks”
- ▶ “in more ideologically extreme social networks, users’ outrage expression behavior is less sensitive to social feedback.”

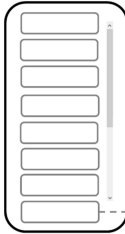
Finding are consistent with reinforcement learning and norm learning, but there are limits to what they can learn just from watching without controlling the environment

## Scrolling Stage

### Outrage Norm Condition



### Neutral Norm Condition



Absolutely disgusting. Enlisting to serve this country should immediately grant citizenship, period. Military veterans ESPECIALLY should NEVER face the threat of deportation.

309 Retweets 221 Likes



Thankful that when I was in the deportation voluntary leave was an option. Even more thankfully, they decided in my favor.

24 Retweets 31 Likes

## Learning Stage

### Choose Tweet

It's a symbol of hate because Trump's racist, hateful supporters are murdering Americans, stupid. The El Paso murderer liked hate photos #GunControl



Let's put our money where our hearts are. Show the el paso survivors who are speaking out support. Contribute \$3 #GunControlNow



30 trials

### Receive Feedback

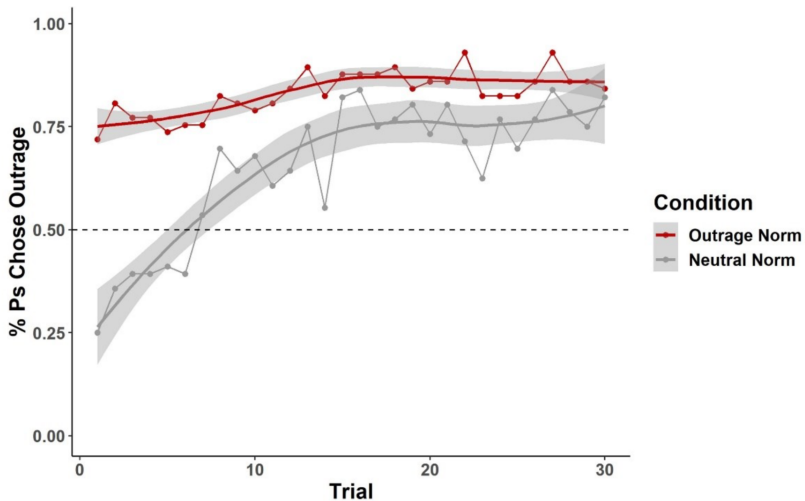


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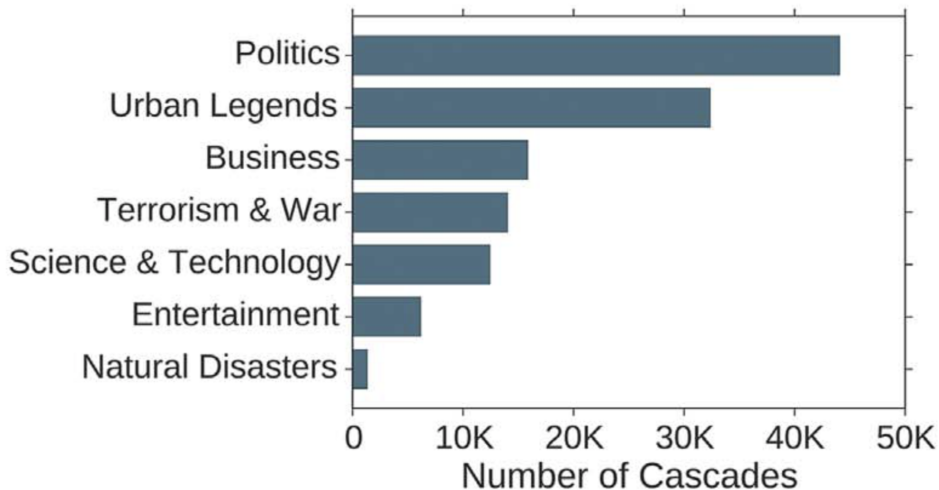
- Now researchers can vary the environment (norms) and feedback (reinforcement)



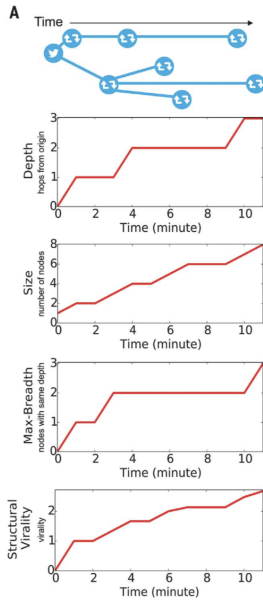
► Norms matter and people learn to choose outrage

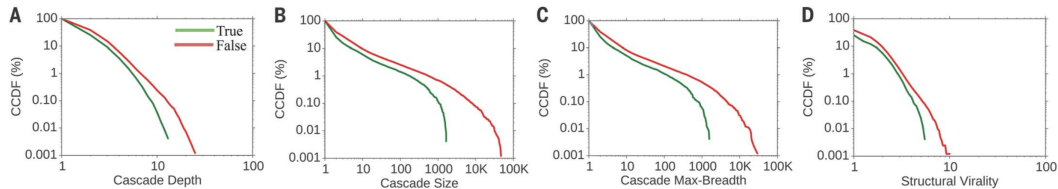
# The spread of true and false news online

Soroush Vosoughi,<sup>1</sup> Deb Roy,<sup>1</sup> Sinan Aral<sup>2\*</sup>



Measured as true or false based on 6 fact checking websites





- ▶ false rumors spread deeper, are more retweeted, spread more broadly, and are more viral than true rumors

What might explain this pattern?

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What might explain this pattern? Voutsourghi et al argue that false rumors are more novel than true rumors and so people decide to retweet them. They don't mention moral outrage.

## Robustness checks

- ▶ Is this because the fact checked rumors are somehow different?

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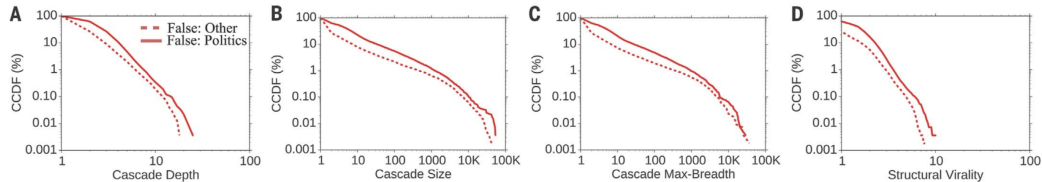
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## Robustness checks

- ▶ Is this because the fact checked rumors are somehow different? No. Newly, independently checked rumors show similar pattern.
- ▶ Is this because of bots? No. If you remove bots you get similar patterns



- ▶ political rumors spreads deeper, are more retweeted, spread more broadly, and are more viral than true rumors

All of this might make you think that we are awash in false rumors about politics, but.

. . . .

# Fake news on Twitter during the 2016 U.S. presidential election

Nir Grinberg<sup>1,2\*</sup>, Kenneth Joseph<sup>3\*</sup>, Lisa Friedland<sup>1\*</sup>,  
Briony Swire-Thompson<sup>1,2</sup>, David Lazer<sup>1,2†</sup>

The spread of fake news on social media became a public concern in the United States after the 2016 presidential election. We examined exposure to and sharing of fake news by registered voters on Twitter and found that engagement with fake news sources was extremely concentrated. Only 1% of individuals accounted for 80% of fake news source exposures, and 0.1% accounted for nearly 80% of fake news sources shared. Individuals most likely to engage with fake news sources were conservative leaning, older, and highly engaged with political news. A cluster of fake news sources shared overlapping audiences on the extreme right, but for people across the political spectrum, most political news exposure still came from mainstream media outlets.

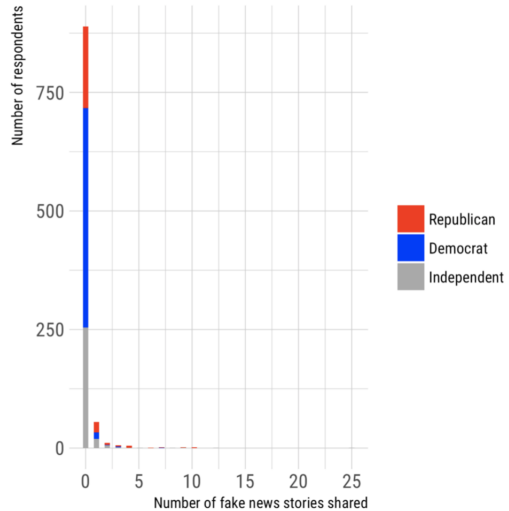
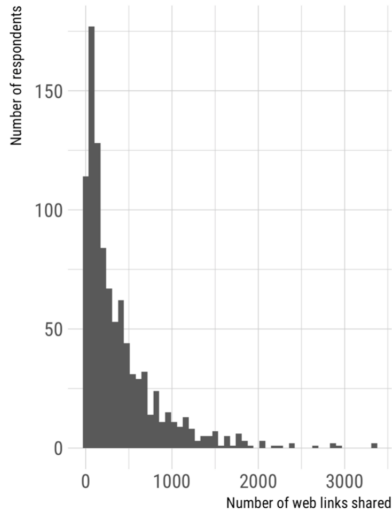
Twitter, 2016 US election, <https://dx.doi.org/10.1126/science.aau2706>

# Less than you think: Prevalence and predictors of fake news dissemination on Facebook

Andrew Guess<sup>1\*</sup>, Jonathan Nagler<sup>2</sup>, Joshua Tucker<sup>2</sup>

So-called “fake news” has renewed concerns about the prevalence and effects of misinformation in political campaigns. Given the potential for widespread dissemination of this material, we examine the individual-level characteristics associated with sharing false articles during the 2016 U.S. presidential campaign. To do so, we uniquely link an original survey with respondents’ sharing activity as recorded in Facebook profile data. First and foremost, we find that sharing this content was a relatively rare activity. Conservatives were more likely to share articles from fake news domains, which in 2016 were largely pro-Trump in orientation, than liberals or moderates. We also find a strong age effect, which persists after controlling for partisanship and ideology: On average, users over 65 shared nearly seven times as many articles from fake news domains as the youngest age group.

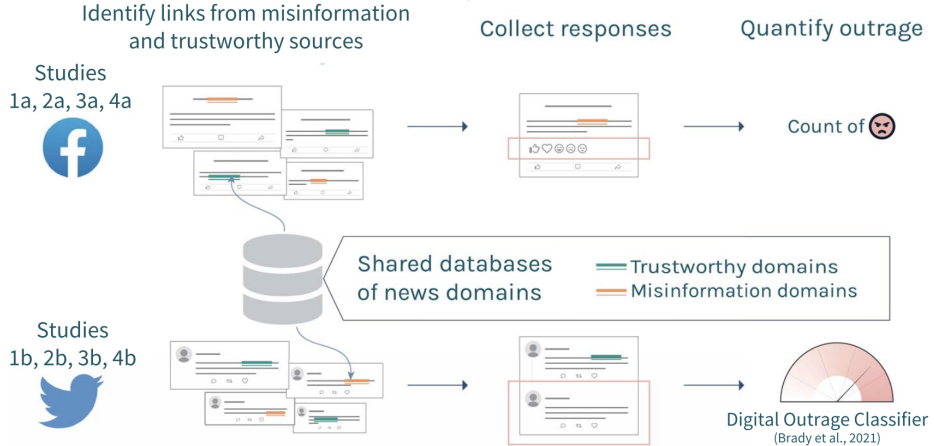
Facebook, 2016 US election, <https://dx.doi.org/10.1126/sciadv.aau4586>



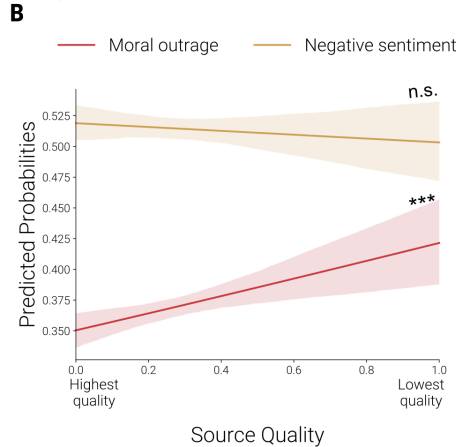
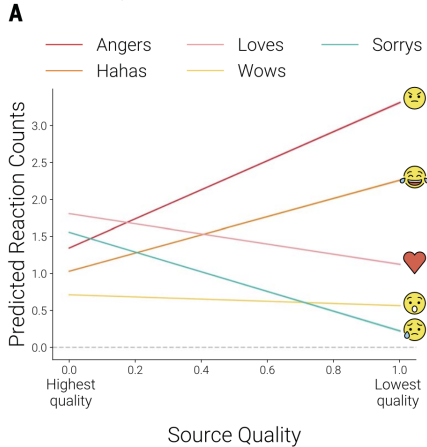
► people share links of Facebook, just not fake news

# Misinformation exploits outrage to spread online

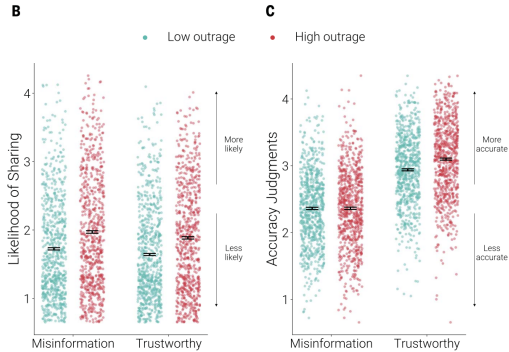
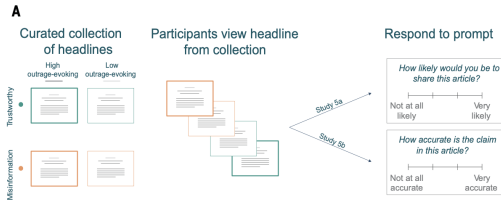
Killian L. McLoughlin<sup>1,2†</sup>, William J. Brady<sup>3\*†</sup>, Aden Goolsbee<sup>4</sup>, Ben Kaiser<sup>5</sup>,  
Kate Klonick<sup>6,7,8,9</sup>, M. J. Crockett<sup>1,5,10\*</sup>



- important to study more than one platform



- low quality sources are more likely to evoke outrage on Facebook (A) and Twitter (B)



► Higher outrage increases sharing but does not affect accuracy discernment

Why do these studies matter? (Vosoughi et al and McLoughlin et al)

- ▶ They impact which policy you might use to intervene (eg, bot removal; focus on epistemic vs nonepistemic motives for sharing)

Stepping back:

- ▶ Social media seems to amplify moral outrage and false rumors

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- ▶ Social media seems to amplify moral outrage and false rumors
- ▶ This happens because of the choices of people and the choices of social media architects

Next class: Social media and democracy